

Prompt History — NM-LAB-08252026 (AI-Assisted Dashboard Generation)

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Tool used: Grok (xAI)

Purpose: Record of prompts issued to the AI assistant, in order, during generation of NM-LAB-08252026_Dashboard.html and its supporting Python pipeline.

Prompt 1 — Task setup and data source specification

You are an expert Python numerical methods engineer and dashboard developer. Solve the laboratory activity NM-LAB-08252026 exactly as specified.

DATA SOURCE - Excel file "Data 01.xlsx", sheet "Sensor Log".

Columns: Reading, Date, Time, Depth (m). Ignore the broken Timestamp column. Construct proper datetime from Date + Time. Convert to numeric axis t = hours elapsed from the first reading (step = 0.25 h). Use all 288 readings (3 full days). The lab text mentions "96" from the previous activity; this continuation uses the full stage log.

Prompt 2 — Tab Group 1 specification (finite-difference derivatives)

TAB GROUP 1 – Finite-difference derivatives

Forward difference at first point, backward at last point, central differences elsewhere for dh/dt (m/h) and d^2h/dt^2 (m/h²). Report time of maximum first derivative and its value (m/h). One-sentence interpretation of the second derivative regarding inflow.

Prompt 3 — Tab Group 2 specification (Levenberg–Marquardt curve fit)

TAB GROUP 2 – Curve fit (Levenberg-Marquardt only)

Choose and defend a model in two sentences. Preferred candidates (implement at least one, preferably the four-parameter logistic or sum of two logistics): Four-parameter logistic: $h = c + a / (1 + \exp(-k \cdot (t - t_0)))$ Gompertz, sum of two logistics, or cubic as baseline.

Initial guess p_0 must be justified from visual inspection of the data (ceiling, inflection, etc.). Never use default ones. Fit with `scipy.optimize.curve_fit(..., method="lm", maxfev=20000)`. Compute in exact order: residuals, SSE, SST, R^2 , $s = \sqrt{\text{SSE}/(n-p)}$, $\text{SE}(b_j) = \sqrt{\text{diag}(\text{pcov})}$, t-stat, two-tailed p-value via `scipy.stats.t`. Report every parameter with SE, t, p-value. Round sensibly (logger resolution is 0.01 m). Residual diagnostics (must answer in text): centred on zero or drift? long runs of same sign? spread widens with level? largest residual vs 1 cm logger resolution?

Prompt 4 — Tab Group 3 specification (numerical integration)

TAB GROUP 3 – Area under the fitted curve

$A = \int_{t_0}^{t_n} \hat{h}(t) dt$ using `scipy.integrate.quad` (not raw points). Report value + absolute error estimate + units ($\text{m}\cdot\text{h}$). Cross-check with `np.trapezoid` on the raw readings. Explain the difference in one sentence. Interpret what the number means (and does not mean) for the flood-control office.

Prompt 5 — Dashboard assembly specification

Single self-contained HTML dashboard

Python script writes ALL results (numbers, arrays, model equation, residual text, conclusions) into the HTML as an embedded JSON block or direct string substitution. No computation in JavaScript. RULE 0: Python computes, HTML only displays. Layout: Header (always visible): raw stage log time series. Tab 1: first & second finite-difference derivatives; mark max dh/dt . Tab 2: fitted curve over raw data + model equation; residual plot (time & vs $\hat{h}$) with zero line; parameter table (value $\pm$ SE, t, p); SSE (with n,p), R^2 , SST, s (with dof), conclusions from p-values and residual pattern. Tab 3: fitted curve shaded to x-axis between limits, integral value, units, trapezoid cross-check, interpretation. Use Chart.js or pure SVG/Canvas from a local copy or CDN (keep offline-capable if possible). One double-click HTML file, no server, no build step. Professional, clean, readable styling. State course/section/issued date/author in footer.

Prompt 6 — Request for underlying Python script

give me the python script for this html

Prompt 7 — Request to expand detail and add identity metadata

make it more detailed with name section and generated by an have complete details and statistics

Prompt 8 — Assistant asked for the student's name; user provided it

Assistant asked: "You haven't told me your name yet. Reply with the name (and any student ID / section if you want them shown)..."

User reply:

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Prompt 9 — Second request for the final Python script

Give me the python script

Prompt 10 — Debugging prompt issued to VS Code AI agent

Tool used: VS Code agent (e.g. Copilot/Claude Code extension)

fix the error and the problem

Prompt 11 — Rename output file, issued to VS Code AI agent

Tool used: VS Code agent (e.g. Copilot/Claude Code extension)

change name of html to "lab0_Lopez.html"

Prompt 12 — Restated lab objective (4-parameter logistic, phase framing, RMSE/MAE, quad integration)

The primary objective of Laboratory Activity 02 is to perform a numerical analysis of a 96-reading reservoir stage log by fitting a 4-parameter logistic model—defined by its upper asymptote (A), growth rate coefficient (B), baseline stage (C), and inflection time (t_0)—to model reservoir behavior across four distinct growth phases: Lag, Exponential Acceleration, Deceleration, and Saturation. Using finite-difference approximations, the first derivative ($\frac{dS}{dt}$) isolates the peak stage increase rate at the inflection point, while the second derivative ($\frac{d^2S}{dt^2}$) tracks inflow acceleration dynamics across the raw time series. Model accuracy is evaluated through point-by-point residual diagnostics ($S_{\text{raw}, i} - S_{\text{fitted}}(t_i)$) and global summary metrics, including Sum of Squared Errors (SSE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Finally, standard units are strictly maintained across all measurements—meters (m) for stage, hours (hr) for time, m/hr and m/hr^2 for derivatives—culminating in the numerical integration ($\int S_{\text{fitted}}(t) dt$) using `scipy.integrate.quad` to calculate the total cumulative area under the fitted level in meter-hours ($\text{m} \cdot \text{hr}$).

Prompt 13 — Ask for missing Residuals-vs-Fitted chart

where is the Residuals against the fitted value, if missing add it

Prompt 14 — Request to make it more prominent

make it visible or expand it

Prompt 15 — Request to compare finite-difference derivatives against the fitted model's smoothed derivative

Add finite differences vs. the smoothed derivative from the fit, both derivatives